Siheng Ye

Electrical and Electronic Engineering | Robotics, Control and Intelligent Systems
Ningbo, China | Portfolio | LinkedIn | GitHub | ORCID

Education

University of Nottingham Ningbo China Expected July 2027
BEng (Hons), Electrical and Electronic Engineering
Research interests: Embedded perception, robotic systems and control, cyber-physical systems, and edge AI.
Current Year 4 modules: Internet of Things (EEEE3124); Control Systems Design (EEEE3066); Robotics, Dynamics and Control (EEEE3076); Sensing Systems and Signal Processing (EEEE3110).

Research Experience

Robot–Elevator Interaction June–August 2026
Lift Robot project | Supervised by David Chieng

- Developed a ROS 2 perception-to-action pipeline on Jetson AGX Orin, combining YOLO11n button detection, HOG+SVM symbol recognition, Qwen3-VL semantic interpretation, RGB-D localisation, and camera-to-base calibration.
- Evaluated offline target selection on 475 requests: 404 correct (85.1%); supported two staged button-activation demonstrations with guarded arm motion.

Raspberry Pi 5 Engineering Assistant June–August 2026
FoSE Undergraduate Research Practice | Supervised by Ian Pratt

- Built a compact-LLM assistant using four-bit Qwen3-1.7B, course-material retrieval, deterministic calculation tools, and a web interface exposing answers, sources, and tool traces.
- Compared seven configurations on 96 EEE questions in an offline GPU-reference evaluation; tool-augmented RAG achieved a mean rubric score of 0.777/1.000.

Bangbang V2: Inclusive Teleoperation Robot March–May 2026
Supervised by David Chieng; co-supervised by Luke Younghoon Chang

- Developed facial-expression recognition paths sharing MediaPipe's 478-landmark front end, including CNN/ONNX and interpretable FACS/action-unit routes.
- Tested the live Pi 5 camera path and contributed to bilingual speech and operator-interface integration; AI outputs remained advisory to the human operator.

Transformable Land–Air Robot June–August 2025
Supervised by Chiew-Foong Kwong; co-supervised by Fuhua Jia

- Contributed to mechanical design, cross-modal motion control, and embedded debugging within the FURP 2025 team, developing a prototype combining ground travel and quadrotor flight.

Selected Engineering Projects

Closed-Loop Forward Converter March–May 2026
EEEE2074 Energy Project | Group 07

- Contributed to a team project designing a 50 W two-switch forward converter with Type III compensation, PLECS parasitic modelling, and investigation of switching-voltage spikes.

Doppler Radar Speed Detector October–November 2025
Embedded signal processing and digital logic project | Group 02

- Contributed to a team prototype integrating HB100 radar, analogue conditioning, STM32U575 acquisition, CMSIS-DSP FFT processing, and a CPLD display with one-way RS485 transfer.

Robotic Vehicle Design October 2024–March 2025
EEEE1039 Applied EEE Construction Project | Group 18

- Contributed PID line-following, OpenCV colour recognition, wireless communication, PWM motor control, and Arduino/Raspberry Pi integration within a four-person team.

Accepted Conference Contributions

S. Ye, Z. Tong, S. Zhang and D. Chieng. "Robot–Elevator Interaction: From Physical Panel Recognition to Activation" (Paper 105). 2026.

Accepted for oral presentation, IHCI 2026.

S. Ye, X. Li, C. Qi, L. Yin, Y. Zhang and I. Pratt. "A Lightweight Test and Training Platform Utilizing a Raspberry Pi 5: AI-retrieval, Tools and a Compact LLM" (Paper 106). 2026.

Accepted for oral presentation, IHCI 2026.

Y. Chang, **S. Ye**, Y. Cao, et al. "Bangbang V2: On-Device Facial Emotion and Bilingual Speech Recognition for a Raspberry Pi-Based Tele-Operated Avatar Robot" (Paper 230). 2026.

Accepted for presentation, HRFEST 2026; camera-ready manuscript in progress.

Technical Skills

Programming: Python, C++, Verilog; MATLAB/Simulink; Linux and SSH.

Robotics and embedded systems: ROS 2, Raspberry Pi, Jetson AGX Orin, STM32, Arduino, PID control, RGB-D localisation, and sensor integration.

Machine learning: OpenCV, MediaPipe, ONNX, RAG, supervised fine-tuning, GGUF/llama.cpp deployment, and comparative evaluation.

Electronics and tools: PLECS, LTspice, Multisim, Git, oscilloscopes, and LCR measurement.

Languages

Mandarin Chinese: Native.

English: IELTS Academic 7.0 (July 2026).

French: Around B1 (self-assessed).